

prepare_model_fold_input.R

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prepare_model_fold_input
<i>Prepare Train and Test Model Inputs for One Fold</i>

Description

Prepares one model fold without using held-out responses or predictors to learn taxon filtering or predictor scaling. Spatial predictors must already represent the training-only construction and held-out projection paths.

Usage

```
prepare_model_fold_input(  
  data_community_matrix = NULL,  
  data_abiotic_wide = NULL,  
  data_spatial_train = NULL,  
  data_spatial_test = NULL,  
  train_ids = NULL,  
  test_ids = NULL,  
  error_family = NULL,  
  min_n_taxa = NULL,  
  age_scale_mode = "z_score"  
)
```

Arguments

- `data_community_matrix`
Numeric community matrix with named sample rows and taxon columns.
- `data_abiotic_wide`
Wide abiotic data frame with 'dataset_name', 'age', and predictor columns.
- `data_spatial_train, data_spatial_test`
Optional raw spatial predictor data frames for training and test samples. Both must be supplied together and have sample identifiers as row names.

<code>train_ids, test_ids</code>	Disjoint character vectors of training and test sample identifiers.
<code>error_family</code>	Character scalar naming the model error family. Binomial responses are converted to presence-absence before training-only taxon filtering.
<code>min_n_taxa</code>	Positive integer giving the minimum retained training taxa.
<code>age_scale_mode</code>	Character scalar passed to <code>'scale_abiotic_for_fit()'</code> . Supported values are <code>"z_score"</code> and <code>"center"</code> .

Value

Named list containing assembled training input, scaled test predictors, aligned test observations, an explicit taxon mapping, and fold diagnostics.

Examples

```
data_community <-
  base::matrix(
    base::c(0, 1, 0, 1),
    ncol = 1,
    dimnames = base::list(
      base::c("a__0", "b__0", "c__0", "d__0"),
      "taxon_a"
    )
  )
data_abiotic <-
  base::data.frame(
    dataset_name = base::c("a", "b", "c", "d"),
    age = 0,
    temperature = base::c(8, 10, 12, 14)
  )
prepare_model_fold_input(
  data_community_matrix = data_community,
  data_abiotic_wide = data_abiotic,
  train_ids = base::c("a__0", "b__0", "c__0"),
  test_ids = "d__0",
  error_family = "binomial",
  min_n_taxa = 1L,
  age_scale_mode = "center"
)
```

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